



E/I/I test tables

Rejected take-off and runway overrun, Tarlton Springs

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INVESTIGATION WORKFLOW SUITE

E/I/I test tables: Rejected take-off and runway overrun, Tarlton Springs

Existence: did it exist or occur? Influence: did it influence this occurrence? Importance: is it worth addressing regardless?

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	AS-139 10.4.2 requires a documented wildlife hazard management plan only where the aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs' heaviest scheduled aeroplane is 22,900 kg and it handled 61,400 passenger movements in 2025, so neither limb applies. Under 10.4.3 its measures need not be documented at all. The aerodrome accordingly held no plan, no hazard assessment, no dispersal programme and no record of measures beyond the single pre-first-arrival inspection at Aerodrome Manual 6.3. Confirmed against the certified aerodrome register and the aerodrome operator's own compliance file; the aerodrome is not in breach.
Influence	Pass - likely (70%)	The absence of a documented plan is the absence of the one mechanism that would have connected the aerodrome's own sighting log to the operators using the aerodrome, and the one that would have put the failed fence on a funded programme rather than an unfunded works register.
Importance	Not tested	(none)

Investigator notes: The aerodrome is not in breach of anything. That is what makes this an External Influences object rather than a compliance finding: the standard's own threshold is what leaves an aerodrome of this size with an undocumented, unaudited and in practice unfunded wildlife programme.

Organisational Influences

Evening service approved as a schedule change, route risk assessment not reopened

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	RRA-041 issue 3 (11/2022) is the current route risk assessment for Tarlton Springs. It carries the operating restriction 'daylight operations only pending review of night lighting and diversion fuel policy'. CMR-2025-118 (approved by the Head of Flying Operations on 22/05/2025) introduced a scheduled evening rotation from 14/07/2025, the first company operation into Tarlton Springs after last light. It assessed the change as 'schedule change only', recorded 'no new hazard identified', and answered 'No' to route risk assessment review required. RRA-041 was not reopened and no issue 4 exists. The Safety Department's records show no involvement beyond receipt of the approved record.
Influence	Pass - likely (85%)	The change put a company aircraft into an aerodrome after last light for the first time, in explicit contradiction of a live restriction, and did so through a process that asked whether the change was new rather than whether the assessment still held.

Importance	Not tested	(none)
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Investigator notes: The clearest single organisational finding in the investigation. The restriction that should have blocked this change was sitting in the operator's own current route risk assessment, and the change record does not mention it.

Perimeter fence repair deferred for three budget cycles

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Works register item WR-221, raised 04/2023: four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, with animals observed passing through, replacement estimated at \$41,000. Deferred at the 2023, 2024 and 2025 budget rounds, each entry reading 'defer, unfunded, review next cycle'. WR-263 (02/2026) records the same section deteriorating further. The investigator's site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark, 110 m before the point at which the first carcass was recovered.
Influence	Pass - very likely (90%)	This is the physical route by which the animals reached the movement area. The pad through the fence at 1,050 m and the carcasses at 1,188 m and 1,221 m are 140 m apart on the same side of the runway.
Importance	Not tested	(none)

Investigator notes: The deferrals are recorded in the aerodrome operator's own register in its own words, including 'animals observed passing through' in the original 2023 entry. Nothing here had to be reconstructed.

Risk Controls

Cabin floor proximity emergency lighting partially unserviceable

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Raised by the first officer, who noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. The functional check on 24/06/2026 found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed, traced to a failed common ballast. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs and neither illuminates nor monitors the floor segments, so the defect was undetectable by the crew or by a line engineer without a dedicated functional test. The last such test was 11 months before the occurrence, inside the 24-month programme interval.
Influence	Fail - unlikely (10%)	A partially dark escape path is a defence that is not there when it is needed, and nothing available to the crew would have told them.
Importance	Pass - very likely (90%)	It did not influence this occurrence and it is still a finding. The disembarkation was unhurried, by the forward airstair, with the main cabin lights on. Change one thing - a fire indication, smoke, a night evacuation over the wing - and two dark segments in the aft cabin are the difference between an orderly exit and a blocked one. The gap that matters is the monitoring: a 24-month interval on a system the daily test cannot see means the fleet's true state is unknown, and the same ballast is fitted across the type.

Investigator notes: Left in draft deliberately. The fleet-wide question is out with engineering and the object should not be finalised until that answer is in, even though its own tests are settled. No

arrow, deliberately: Influence failed at 10%. It did not contribute to the overrun or to its outcome, because no evacuation was commanded and the main cabin lights were on throughout; it is carried on Importance alone, as a finding about a defence that was not there.

Rejected take-off training covers aircraft system failures below V1 only

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	TRG-SYL-07 module 4's four scripted scenarios are engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt continued. All four are aircraft system events and three are below V1. No crew has been presented with an external obstruction scenario in the five training cycles on file, covering 214 crew in the current cycle alone. Part B 4.6 matches: 4.6.3 lists reject items to V1, 4.6.4 states the take-off shall be continued at or above V1, and the section contains no guidance on an obstruction on the runway at any speed. Both the pilot in command and the first officer stated independently that every stop or go scenario they have been given has been an engine and has arrived below V1.
Influence	Pass - likely (75%)	The captain had two seconds and nothing behind him. The decision he made was the first time he had considered the question at all, which is what the training system is supposed to prevent.
Importance	Not tested	(none)

Investigator notes: Both crew raised this unprompted, in almost the same words, and the syllabus bears them out. Worth being careful in the report that this explains the decision rather than excusing it.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Operations Manual Part A 8.14.2 requires the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to commence the roll until the result is passed. 8.14.3 requires nothing at other aerodromes. Appendix 8A lists an aerodrome where a wildlife strike involving a company aircraft has been reported there in the preceding 24 months. Nine aerodromes are listed; Tarlton Springs never has been. No provision anywhere in Part A asks a crew to report a sighting, and no provision admits an aerodrome to the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.
Influence	Pass - very likely (90%)	This is the operator's own defence against exactly this occurrence, and it was unarmed. Eleven sightings of kangaroos on the sealed surface at this aerodrome could not put it on the list, because they were not strikes and were not reported to the company.
Importance	Not tested	(none)

Investigator notes: The safety issue the map exists to find. The control is sound and the trigger is circular: it can only be armed by the event it is there to prevent. Recommend this one in its own right regardless of anything else in the report.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

Test	Result	Supporting text
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Existence	Pass - extremely likely (95%)	The roll began at 17:30 local, 32 minutes after sunset and 5 minutes after the end of civil twilight, with moonrise not until 22:41 and the moon 14% illuminated. The aerodrome was in full darkness with no natural ambient light on the strip and no town lighting within sight. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip either side of the sealed surface is not visible from the centreline. The aerodrome's own sighting log records 38 of its 41 kangaroo entries between one hour before and two hours after last light.
Influence	Pass - likely (80%)	The occurrence happened inside the two-hour window in which this aerodrome's own records show almost all of its wildlife movement, and in the light condition in which nothing off the sealed surface can be seen.
Importance	Not tested	(none)

Investigator notes: The condition that connects the schedule change to the animals. On its own it is unremarkable; it is what makes the missing inspection matter.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Sign-on 06:30 local, five sectors, the fifth added that morning when the rostered crew reported unfit. The occurrence was at 17:30 local, 11 hours 00 minutes into a duty planned to finish at 18:35, giving 12 hours 05 against a limit of 12 hours 30 for five sectors from that sign-on. The aircraft departed 40 minutes behind schedule. Both crew described themselves in interview as tired but not struggling; the captain said he would have declined a further sector and that 'for the two seconds that mattered, I was there'.
Influence	Fail - unlikely (25%)	A late fifth sector at the end of a long duty into an aerodrome with no alternate at night is a routine exposure on this schedule, not a one-off.
Importance	Pass - likely (85%)	It failed Influence for this occurrence and remains a finding worth acting on. The evening Tarlton Springs rotation routinely places the last sector inside the last hour of a 12-hour duty, at an aerodrome with no company engineering, no alternate within 200 nm at night, and no crewing resilience, so a single unfit crew member turns a four-sector day into a five-sector one on the morning of operation. Nothing about that changes because it did not bite here.

Investigator notes: The tempting object. It exists, it reads as a cause, and the evidence will not carry it for this occurrence. Keeping it as an other risk factor is the honest answer and it is still going in the report. No arrow, deliberately: Influence failed at 25%, so every mechanism fatigue would have had to work through is contradicted by the recorded data, and an arrow would assert a causal path this record does not have. It did not contribute to the overrun; it is a finding about the schedule.

Runway edge lighting intensity reduced the crew's ability to detect the animals

Test	Result	Supporting text
Existence	Inconclusive - about as likely as not (45%)	For: the pilot in command recalled the lighting as 'brighter than I'd want', a corridor of light with everything outside it black, and said anything off the seal could not be seen coming. A photometric check seven days later, on 25/06/2026, put four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3, in the section the animals crossed, and the aerodrome's maintenance programme checks lamp serviceability without ever measuring intensity. The intensity deviation is settled; the step from that reading to a

		reduced ability to see an animal on the runway 09 surface is an inference, and the inference is what this object tests.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Left in draft on purpose. Both accounts are honest and they do not agree, the one measurement shows a deviation but not an effect, and there is no way to close it with what is available. Say so in the report rather than picking a side.

Take-off commenced with a tailwind above limits

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a stop that runs 62 m past the end invites a question about the wind. It fails. The 06:30, 07:00 and 08:00 UTC observations give 090 at 7, 090 at 6 and 090 at 5 kt, a headwind component of 5 to 7 kt on runway 09. Recorded FDR wind during the roll gives a headwind component of 4 to 6 kt throughout, with no tailwind component at any point. The crew's performance entry used a 5 kt headwind, which matches. There is no evidence of a tailwind at any stage of the roll, from any source.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Raised early because the overrun distance invited it. Disposed of in an afternoon and kept on the map so the report can say it was tested. No arrow, deliberately: Existence failed, so there is no event here to draw an arrow from. The wind did not contribute to the overrun because there was no tailwind at any point in the roll.

Individual Actions

Maximum braking and No. 1 reverse applied without delay

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Power levers reached flight idle 0.6 seconds after the decision and maximum braking was established at 07:30:34, 2 seconds after it, against a recorded fleet mean of 3.2 seconds from decision to maximum braking. Anti-skid was active on all four wheels throughout and brake energy reached 84% of the certified maximum without a brake fire. Reverse was selected on No. 1 within 4 seconds; No. 2 was unavailable because its propeller had been destroyed in the strike. The first officer made the speed callouts and the runway-end call. The aircraft left the seal at 21 kt, stopped 62 m into a 90 m RESA, upright, on the extended centreline, and was disembarked without an evacuation.
Influence	Pass - very likely (90%)	The overrun was 62 m into a prepared surface at low speed rather than a departure from the RESA at speed. The margin between those two outcomes is the two seconds and the asymmetric braking technique.
Importance	Not tested	(none)

Investigator notes: Recorded as a positive event because it is one. The same investigation that finds the reject was outside procedure should say that what followed it was executed better than the fleet average.

No runway inspection requested before the first departure after last light

Test	Result	Supporting text
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Existence	Pass - virtually certain (100%)	Neither the CVR nor the tower's recorded audio contains any request for a runway inspection, any offer of one, or any mention of wildlife at any point before departure. The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local before the first arrival, result 'movement area clear', with none afterwards and none requested. Aerodrome Manual 6.3 provides for inspections outside the scheduled time on the request of the reporting officer or a pilot in command, and the reporting officer confirmed one would have been carried out if asked. Both crew stated they did not consider it; the captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not.
Influence	Pass - likely (85%)	The one action that would have put a vehicle on the runway in the 20 minutes before the roll was available, costs nothing, and was never on either crew member's mind because no document put it there.
Importance	Not tested	(none)

Investigator notes: Careful with the wording of this one in the report. It is an omission with no owner: nothing asked either crew member to consider it, and the aerodrome would have done it if asked.

Take-off rejected 6 kt above V1

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The FDR records V1 called at 112 kt at 07:30:29 and both power levers retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots past V1. Part B 4.6.4 states the take-off shall be continued at or above V1. The CVR records the whole decision in one exchange: the first officer's sighting call at 07:30:31, 'Stopping, I have control' at 07:30:32. The captain's own account, in interview and in his air safety report, is that he judged rotating into two large animals would leave him airborne at night with unassessable damage over country with no usable option, and that he took the risk whose end he could see. He states plainly that it was not by the book.
Influence	Pass - extremely likely (95%)	Rejecting at 118 kt rather than continuing is the decision that put the aircraft into a stop it did not have the runway for. The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1.
Importance	Not tested	(none)

Investigator notes: Do not let this one carry the report. It is a contributing factor and it is downstream of the training object and the animals being there at all.

Occurrence Events

Deceleration achieved below the scheduled accelerate-stop performance

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m, assuming the stop begins at or before V1 with symmetric reverse. Distance reconstruction from recorded groundspeed puts the aircraft 1,152 m along the runway at the reject, leaving 695 m of sealed surface, and the stopping distance actually achieved from 118 kt was 757 m. The mean deceleration from maximum braking to the runway end was 2.6 m/s ² , approximately 15% below the value used in the certified data for this weight and condition, with reverse available on No. 1 only.

Influence	Pass - virtually certain (100%)	62 m of overrun is the arithmetic of a stop begun 6 kt late with 15% less deceleration than the figures assume. Both contributions are needed to reach the end; neither alone does.
Importance	Not tested	(none)

Investigator notes: The convergence point. Both chains arrive here, which is why the report's analysis should treat it as one mechanism rather than two separate findings.

No. 2 engine power loss during the take-off roll

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a reject at 118 kt is most commonly an engine event. It fails. The FDR records both engines at 98.4% and 98.5% torque, within 0.1% of each other, from power set at 07:30:02 until the power levers were retarded at 07:30:32, with fuel flow, ITT and propeller RPM on both engines inside normal scatter throughout. No caution, warning or exceedance is recorded at any point before the reject. The CVR contains no engine-related call: the captain's first words are 'Stopping, I have control' one second after the first officer's sighting call. The borescope of both engines found no pre-existing damage. The No. 2 propeller damage and the RPM decay both begin at 07:30:33, after the reject and at the moment of impact.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: The first hypothesis anyone offers for a high-speed reject, and the first one the data kills. Kept on the map because the report should show it was tested, not assumed away. No arrow, deliberately: Existence failed, so there is nothing to draw an arrow from. The No. 2 power loss did not contribute to the reject because it had not happened yet; it begins at impact, one second after the reject.

No. 2 propeller and right main gear struck two kangaroos

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Impact at 07:30:33 at 114 kt, one second after the reject began. The FDR records a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits with one blade tip separated, and animal tissue was recovered from two blade roots. The right main gear doors were distorted and the lower right fuselage skin punctured by propeller debris. No. 2 propeller RPM decays from the moment of impact and the No. 2 reverse interlock did not make.
Influence	Pass - virtually certain (100%)	The strike removed one of the two reverse units two seconds into a stop with a 67 m scheduled margin, and it is why the aircraft went off the end rather than stopping on it.
Importance	Not tested	(none)

Investigator notes: The link to make sure the report draws: this is not just damage, it took away half the reverse thrust in the middle of a stop that had no margin.

Runway overrun onto the runway end safety area

Test	Result	Supporting text
Existence	Pass - virtually	Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, departed the end of runway 09 at Tarlton Springs Aerodrome at approximately 21 kt at

	certain (100%)	07:30:51 UTC on 18/06/2026 and came to rest at 07:30:57, 62 m into the 90 m runway end safety area, upright and on the extended centreline, with the nose wheel in graded gravel. 44 passengers, 2 flight crew and 1 cabin crew were on board. Continuous tyre marks run from 1,160 m to the runway end and the nose wheel rut continues to the stopping point.
Influence	Pass - virtually certain (100%)	Substantial damage: three No. 2 propeller blades beyond limits, a shock-load inspection on the engine, distorted right main gear doors, both right main tyres deflated through the fuse plugs at 84% of certified brake energy, and punctures to the lower right fuselage. Nil serious injuries; one cabin crew member reported bruising during the disembarkation. Two kangaroos killed. The aerodrome closed for 4 hours 20 minutes. The aircraft stopped 28 m short of the end of the prepared safety area; beyond it the terrain is ungraded.
Importance	Not tested	(none)

Investigator notes: The critical event. Everything else on this map exists to explain it.

Two kangaroos entered the runway strip and moved onto the sealed surface

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Two adult kangaroos were recovered from runway 09 at 1,188 m and 1,221 m from the threshold, on the northern half. The first officer saw them from the flight deck crossing from the northern side to the south, estimating a range of around 200 m at first sight, and called them at 07:30:31. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass on the same side. Eleven entries in the aerodrome's sighting log record kangaroos on the sealed surface of runway 09/27, all around last light. No inspection had been carried out since 07:35 local that morning.
Influence	Pass - extremely likely (95%)	Nothing else on this map matters without this. It is also the only object on the map that a single \$41,000 works item would have addressed directly.
Importance	Not tested	(none)

Investigator notes: The event everything upstream is about. Note the carcass positions against the fence pad: the geometry does the arguing here, not the interviews.